

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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STUDENT-TEACHER-SUBJECT CODE DATABASE

Aim:

To implement a Prolog database representing the relationship between students, teachers and subjects.

Objective:

- ☐ **To store student information.**
- ☐ **To associate teachers with subjects.**
- ☐ **To determine which teacher teaches a particular subject.**
- ☐ **To determine which subjects are taken by a student.**

Algorithm:

- ☐ **Start.**
- ☐ **Define student-subject facts.**
- ☐ **Define teacher-subject facts.**
- ☐ **Create rules to relate students, subjects and teachers.**
- ☐ **Query the required relationship.**
- ☐ **Display the result.**
- ☐ **Stop.**

Flowchart:

START

|

v

Store Student-Subject Facts

|

v

Store Teacher-Subject Facts

|

v

Enter Query

|

v

Match Facts/Rules

|

v

Display Result

|

v

STOP

Prolog Program:

% Student-Teacher-Subject Database

student(rahul, ai).

student(rahul, mathematics).

student(priya, ai).

student(priya, physics).

student(arun, mathematics).

student(arun, physics).

teacher(ramesh, ai).

teacher(suresh, mathematics).

teacher(kavitha, physics).

teaches(Teacher, Student, Subject) :-

student(Student, Subject),

teacher(Teacher, Subject).

Output:

```
68 ?- teacher_of(Teacher, c_programming).  
Teacher = kumar.
```

Result:

Thus, the Prolog program successfully represents and queries student-teacher-subject relationships.